$S = \{1, 2, 3, 4, 5\}$

$$\mathbb{P} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 0 & 1/4 & 1/4 & 1/4 \\ 0 & 1/2 & 1/3 & 0 & 1/6 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/3 & 0 & 0 & 1/3 & 1/3 \end{bmatrix} \end{array}$$

Irreducible ✓ Aperiodic ✓

Recurrent + Finite State Space $\Rightarrow$ Positive Recurrent ✓

We will eventually talk about the rate of this convergence.

To this end, we will talk about measuring the "distance" between two probability distributions.

For us, it will be the limiting distribution versus the distribution we see after n steps.

Let X and Y be discrete random variables with probability mass functions

$$f_X(x) = P(X = x) \quad \text{and} \quad f_Y(y) = P(Y = y)$$

<u>Definition</u>: The <u>total variation norm distance</u> between $f_X(x)$ and $f_Y(y)$ is denoted by

$$\| f_X - f_Y \|_{TV}$$

and is defined by

$$\|f_X - f_Y\|_{TV} = \frac{1}{2}\sum_z |f_X(z) - f_Y(z)|$$

<u>Example</u>:

x	1	2	3
$P(X=x)$	$1/3$	$1/3$	$1/3$

$f_X(x)$

y	1	2	3
$P(Y=y)$	$1/2$	$1/4$	$1/4$

$f_Y(y)$

$$\| f_X - f_Y \|_{TV} =$$

$$\tfrac{1}{2}\left[\left|\tfrac{1}{3} - \tfrac{1}{2}\right| + \left|\tfrac{1}{3} - \tfrac{1}{4}\right| + \left|\tfrac{1}{3} - \tfrac{1}{4}\right| \right]$$

$$= \tfrac{1}{2} \cdot \tfrac{1}{3} = \tfrac{1}{6}$$

Example:

x	1	2	3
$P(X=x)$	$1/3$	$1/3$	$1/3$

$f_X(x)$

y	1	2
$P(Y=y)$	$1/2$	$1/2$

$f_Y(y)$

$$P(Y=3)=0$$

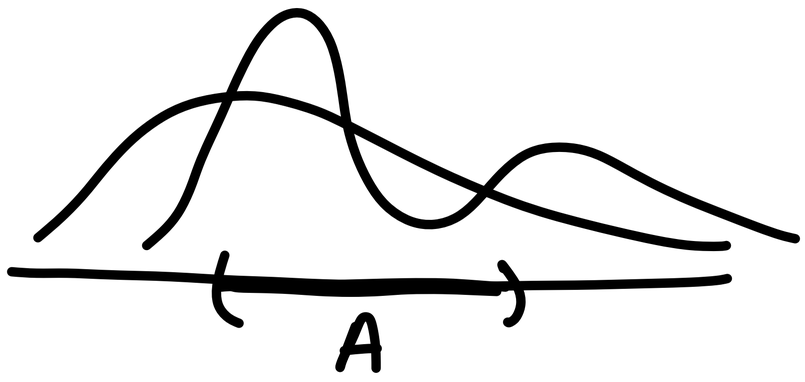
$$\| f_X - f_Y \|_{TV} =$$

$$\tfrac{1}{2}\left[\left|\tfrac{1}{3} - \tfrac{1}{2}\right| + \left|\tfrac{1}{3} - \tfrac{1}{2}\right| + \left|\tfrac{1}{3} - 0\right| \right]$$

$$= \tfrac{1}{2} \cdot \tfrac{2}{3} = \tfrac{1}{3}$$

Why the ½ ?

The total variation norm is defined, more generally, as



Why the ½ ?

The total variation norm is defined, more generally, as

$$\| f_X - f_Y \|_{TV} = \max_A \left| \underbrace{P(X \in A)} - P(Y \in A) \right|$$

defined as an integral if X is continuous

$$\| f_X - f_Y \|_{TV} = \max_A |P(X \in A) - P(Y \in A)|$$

One can show that, for discrete X and Y, this is equal to

$$\frac{1}{2} \sum_z |f_X(z) - f_Y(z)|$$